

[Carnitine: Fact Sheet for Consumers \(original version\)](#).

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What is carnitine and what does it do?

Carnitine is a nutrient found in many foods. Your body also makes carnitine. It helps turn the food you eat into the energy you need.

How much carnitine do I need?

Healthy people's bodies can make all the carnitine they need. Therefore, carnitine is not an essential nutrient, and experts haven't set a recommended daily amount.

What foods provide carnitine?

Many foods, especially animal foods, naturally contain carnitine. These include the following:

- Red meat is a good source of carnitine.
- Poultry, fish, and dairy have some carnitine.
- Vegetables, fruits, and grains have very small amounts of carnitine.

What kinds of carnitine dietary supplements are available?

Some dietary supplements contain carnitine either alone or combined with other ingredients. The two main forms of carnitine in dietary supplements are L-carnitine and acetyl-L-carnitine. Your body absorbs the carnitine from food much better than it absorbs carnitine from dietary supplements.

Am I getting enough carnitine?

Healthy children and adults make all the carnitine they need. However, certain groups of people might have low levels of carnitine because their bodies don't make enough:

- Babies born prematurely
- People with end-stage kidney disease
- People on kidney dialysis
- People with primary carnitine deficiency, a rare genetic disorder

What are some effects of carnitine on health?

Scientists are studying carnitine supplements to understand whether they affect health. Here are some examples of what this research has shown.

Alzheimer's disease and dementia

Studies of carnitine supplements in people with Alzheimer's disease or other forms of dementia have had mixed results. A few studies have shown that carnitine might help people with mild dementia or Alzheimer's disease. Other studies have shown only short-term improvements or no effect at all. More research is needed to understand whether carnitine supplements affect Alzheimer's disease and dementia.

Heart disease and peripheral artery disease

Studies of carnitine supplements for heart disease have had mixed results. Some studies have found that carnitine might reduce the risk of arrhythmias and angina but do not lower the risk of heart failure or heart attack. Other studies have shown that taking carnitine supplements for 6 months or longer might *increase* the risk of heart disease.

Peripheral artery disease is a blood vessel disorder that causes leg pain and cramping. It is common in older people. Some studies have shown that carnitine supplements reduce leg pain during walking, but other studies have found no benefit. More research is needed to better understand the effects of carnitine supplements on heart health and peripheral artery disease.

Insulin resistance and diabetes

Insulin resistance is a condition in which your body doesn't properly use insulin, a hormone that controls blood glucose (also called blood sugar) levels. Insulin resistance increases the risk of type 2 diabetes. Some studies have found that carnitine supplements might help the body use insulin more effectively and might decrease blood glucose and A1C levels. Other studies have shown that carnitine might reduce cholesterol, but not triglyceride, levels in people with diabetes. However, these studies were small. More research is needed to better understand whether carnitine supplements help reduce the risk of insulin resistance and diabetes.

Infertility

Several studies have shown that carnitine supplements improve sperm health in men with infertility. However, carnitine does not appear to increase rates of successful pregnancy in their female partners.

Carnitine supplements have also been studied for infertility in females who have polycystic ovary syndrome (PCOS). One study reported higher rates of ovulation and pregnancy and a lower risk of miscarriage in the individuals who took carnitine supplements plus an infertility drug compared with those who took the drug alone. Another small study found that carnitine supplements improved menstrual period regularity and ovulation and increased the chances of becoming pregnant but did not affect the risk of miscarriage. More research is needed to determine whether carnitine improves infertility or pregnancy rates in people with PCOS.

Osteoarthritis

Studies of carnitine supplements for osteoarthritis have had mixed results. One small study showed that carnitine helped reduce knee pain in middle-age women with osteoarthritis. Another small study in women with obesity and knee arthritis found that carnitine did not improve pain, stiffness, or ease of moving. Larger studies in more diverse groups of people are needed to understand whether carnitine supplements help osteoarthritis symptoms.

Athletic performance

Most of the carnitine in your body is stored in your muscles. Researchers are studying whether carnitine supplements help muscles work more efficiently, but studies examining if carnitine supplements affect athletic performance have had mixed results.

In one small study of young recreational athletes, those who drank a liquid containing carnitine found that it was easier to complete a fitness and endurance test compared with those who drank the liquid without carnitine. In another small study of men, carnitine did not affect performance in a 1-hour cycling exercise, but it did slightly increase the amount of carnitine stored in the muscles of the men who followed vegetarian diets. In other research in trained athletes and active adults, carnitine supplements improved performance and shortened exercise recovery time in some studies but not in others. More research is needed to better understand whether carnitine supplements improve athletic performance.

Weight loss

Because carnitine helps turn the food you eat into the energy you need, researchers have studied whether carnitine supplements can help people lose weight.

Studies of carnitine supplements in middle-age adults with type 2 diabetes have had mixed results. One 6-month study found no effect on weight loss, but another study found that those who took the weight-loss drug orlistat plus carnitine supplements for 1 year lost an average of 4 pounds more than those who only took orlistat. Other studies also found that people who took carnitine supplements lost more weight than people who didn't take them, but the difference was small—only a few pounds. Larger studies are needed to better understand the effects of carnitine supplements on weight loss.

Can carnitine be harmful?

Carnitine in foods and beverages is safe. Therefore, carnitine doesn't have an upper limit. However, taking 3 grams or more a day of carnitine as a dietary supplement can cause nausea, vomiting, stomach cramps, diarrhea, and a fishy body odor. High amounts can also cause muscle weakness in people with chronic kidney disease and can increase the risk of seizures in people who have a seizure disorder.

Does carnitine interact with medications or other dietary supplements?

Some medications can affect carnitine levels in your body. Here are two examples:

- Long-term use of some antibiotics, such as pivampicillin, used to prevent urinary tract infections, can lower the amount of carnitine in your body.
- Anticonvulsant medications, such as phenobarbital and valproic acid, can reduce blood levels of carnitine.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients such as carnitine.

Carnitine and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (e.g., during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](https://www.dietaryguidelines.gov).

Where can I find out more about carnitine?

- For general information on carnitine
 - Office of Dietary Supplements Health Professional Fact Sheet on [Carnitine](#)
- For more advice on choosing dietary supplements
 - [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy diet
 - [Dietary Guidelines for Americans](#)

Glossary

A1C

A blood test that measures the average level of sugar (glucose) in your blood during the past 3 months. An A1C test is used to help diagnose type 2 diabetes. It also shows how well your blood glucose levels have been controlled. Also called a hemoglobin A1C test.

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

Alzheimer's disease

A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

angina

Chest pain caused by a decrease in the amount of blood flowing into the arteries of the heart. The pain can be severe and crushing and may also be felt in the shoulders, arms, neck, jaw, or back. It is caused by a buildup of plaque in the arteries of the heart, which reduces the supply of blood and oxygen reaching the heart. Physical or emotional stress can trigger angina. The pain usually stops within minutes after the stress ends.

antibiotic

A drug used to treat infections caused by bacteria and other microorganisms.

anticonvulsant

A drug that prevents, reduces, or stops convulsions or seizures.

arrhythmia

An irregular heartbeat that can make you feel faint or dizzy because your heart is not pumping enough blood through your body. Arrhythmia is treatable but if not treated it can damage your heart, brain, and other organs and can lead to stroke or heart attack.

arthritis

A group of diseases in which one or more joints (places in the body where two bones connect) become swollen and painful. The most common type of arthritis is osteoarthritis. It is caused by the breakdown of cartilage, a type of tissue that cushions and supports the joint. Without cartilage, the bones in the joint rub together, causing inflammation (swelling, redness, pain, and warmth) and stiffness. Arthritis may affect the fingers, hips, knees, lower back, feet, or any joint in the body.

blood sugar

The main source of energy used by the body's cells. Blood sugar comes from food and is made by the liver, and is carried to the cells through the bloodstream. Also called blood glucose.

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

cholesterol

A substance found throughout the body. It is made by the liver and is an important component of cells. Cholesterol is also used to make hormones, bile acid, and vitamin D. Foods that come from animals contain cholesterol, including eggs, dairy products, meat, poultry and fish. High blood levels of cholesterol increase a person's chance (risk) of developing atherosclerosis and heart disease.

chronic

Happening for a long time, persistently, or repeatedly.

dairy food

Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency

An amount that is not enough; a shortage.

dementia

Damaged brain function (thinking, learning, making decisions, remembering) that worsens over time. It disrupts activities of daily living such as bathing, dressing, and walking.

diabetes

A disease in which blood sugar (glucose) levels are high because the body is unable to use glucose properly. Diabetes occurs when the body does not make enough insulin, which helps the cells use glucose, or when the body no longer responds to insulin.

dialysis

The process of filtering the blood when the kidneys are not able to cleanse it.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

essential

In nutrition, essential nutrients are ones that we must consume for good health because our bodies cannot make them. Essential nutrients include vitamins and minerals.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

gram

g. A measure of weight. It is a metric unit of mass equal to 0.001 kilogram (it weighs 0.035 ounce).

health care provider

A person who supplies health care services. Health care providers include individuals with professional training (including doctors, nurses, technicians, and aides).

heart attack

The blockage of an artery supplying blood and oxygen to the heart, resulting in the damage or death of a section of heart muscle.

heart failure

A condition in which the heart is unable to pump the amount of blood needed by the body. It is caused by high blood pressure, heart attack, and other disorders of the heart or blood vessels. Also called congestive heart failure.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

ingredient

In a dietary supplement, an ingredient is a component of the product, such as the main nutrient (vitamin, mineral, herb, amino acid, or enzyme) or any binder, color, filler flavor, or sweetener. In herbal supplements, the common name and Latin name (the genus and species) of the plant is given in the ingredient list. On a dietary supplement label, the ingredients are listed by weight, with the ingredient used in the largest amount first on the list and the ingredient used in the least amount at the end of the list.

insulin resistance

A condition in which glucose (blood sugar) cannot be absorbed by the cells and used for energy. Instead, glucose builds up in the blood and the body produces more and more insulin (which normally would help glucose get into the cells), resulting in abnormally high blood levels of both glucose and insulin. This can lead to pre-diabetes, type 2 diabetes, and other serious health problems.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

kidney disease

A condition that lessens the ability of the kidneys to filter wastes from the blood, keep blood chemical levels balanced, and make certain hormones. The two most common causes of kidney disease are diabetes and high blood pressure. People with long-term kidney disease may need dialysis or a kidney transplant to stay alive.

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

miscarriage

The natural loss of a fetus before 20 weeks of pregnancy. Also called spontaneous abortion.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

obesity

A condition characterized by an excessive amount of body fat. Obesity can be assessed by calculating the body mass index (BMI). (BMI is a number that estimates the amount of body fat on a person, based on weight and height. In adults, a BMI of 30 or higher indicates obesity. Some people, such as bodybuilders or other athletes with a lot of muscle, can be overweight without having obesity. See: overweight.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

peripheral artery disease

A disorder in which the arteries supplying blood to the kidneys, stomach, arms, legs, or feet become blocked by a build-up of fat and cholesterol deposits inside the arteries (atherosclerosis), causing cramping and weakness.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

polycystic ovary syndrome (PCOS)

An imbalance of hormones that can cause irregular ovulation, irregular menstrual periods or none at all; increased facial or body hair; acne; abnormal growths in the ovaries; weight gain; and infertility. PCOS affects females of child-bearing age and is more common in individuals with obesity or who have a mother or sister with PCOS. Having PCOS increases the risk of high blood pressure, heart disease, type 2 diabetes, and endometrial and ovarian cancer. Also called polycystic ovarian syndrome.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

seizure

Sudden changes in behavior caused by excessive electrical activity in the brain.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

triglyceride

A type of fat found in your blood. When you eat more than you need, your body turns the excess calories into triglycerides. High blood levels of triglycerides can increase your risk of heart disease, heart attack, and stroke.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

urinary tract

The organs involved in making and relieving the body of urine, including the kidneys, ureters, bladder, and urethra.

vegetarianism

The practice of avoiding all or most animal products for environmental, philosophical, and health reasons. Vegetarians (people who practice vegetarianism) eat a diet based on foods that come from plants and may include some dairy products and eggs. See: vegetarian diet.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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